```
pip install Numpy
```

```
Looking in indexes: https://pypi.org/simple, https://us-python.pkg.dev/colab-wheels/public/simple/
Requirement already satisfied: Numpy in /usr/local/lib/python3.7/dist-packages (1.21.6)
```

```
import numpy as np
data = np.random.randn(2,3)
data
```

```
array([[ 1.99679861,  1.20067934, -1.90228427],
       [ 2.1340671 ,  2.10758684,  0.96026382]])
```

```
data *10
```

```
array([[ 19.96798607,  12.00679336, -19.02284271],
       [ 21.34067101,  21.07586835,   9.60263817]])
```

```
data + data
```

```
array([[ 3.99359721,  2.40135867, -3.80456854],
       [ 4.2681342 ,  4.21517367,  1.92052763]])
```

```
data.shape
```

```
(2, 3)
```

```
data.dtype
```

```
dtype('float64')
```

```
data1 = [6, 7.5, 8, 9, 1]
arr1 = np.array(data1)
arr1
```

```
array([6. , 7.5, 8. , 9. , 1. ])
```

```
d2 = [[1,2,3,4],[5,6,7,8]]
arr2 = np.array(d2)
arr2
```

```
array([[1, 2, 3, 4],
       [5, 6, 7, 8]])
```

```
arr2.ndim
```

```
2
```

```
arr2.shape
```

```
(2, 4)
```

```
arr1.dtype
```

```
dtype('float64')
```

```
arr2.dtype
```

```
dtype('int64')
```

```
np.zeros(10)
```

```
array([0., 0., 0., 0., 0., 0., 0., 0., 0., 0.])
```

```
np.zeros((3,6))
```

```
array([[0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0.]])
```

```
np.empty((2,3,2))
```

```
array([[[2.2677866e-316, 0.0000000e+000],
        [0.0000000e+000, 0.0000000e+000],
        [0.0000000e+000, 0.0000000e+000]],

       [[0.0000000e+000, 0.0000000e+000],
        [0.0000000e+000, 0.0000000e+000],
        [0.0000000e+000, 0.0000000e+000]]])
```

```
np.arange(15)
```

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14])
```

```
#type cast
arr = np.array([1,2,3,4,5,6])
arr.dtype
```

```
dtype('int64')
```

```
floatarr=arr.astype(np.float64)
floatarr.dtype
```

```
dtype('float64')
```

```
a=np.arange(10)
a
```

```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
a[5]
```

```
5
```

```
a[5:8]
```

```
array([5, 6, 7])
```

```python
a[5:8]=12
a
```

```
array([ 0,  1,  2,  3,  4, 12, 12, 12,  8,  9])
```

```python
a_slice = a[5:8]
a_slice
```

```
array([12, 12, 12])
```

```python
a_slice[:]=30
a_slice
```

```
array([30, 30, 30])
```

```python
a
```

```
array([ 0,  1,  2,  3,  4, 30, 30, 30,  8,  9])
```